

SOLUTION REQUIREMENTS

FitBuddy AI Fitness Plan Generator - Gemini Models

Date	22.09.2026
Team ID	SWTID-2026-5226
Project Name	Fitbuddy AI Fitness Plan Genetor Gemini Modls
Maximum Marks	4 Marks

Team Members

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Step 1: Functional Requirements (FR)

Functional requirements define the specific behaviors, functions, and features the FitBuddy system should provide.

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users should be able to securely sign in and access their personal fitness profile.	High
2	Authorization levels	Normal users can manage their own plans and progress; an administrator can manage system settings and user access.	Medium
3	External interfaces	The system should connect to Gemini models through an API to process user inputs and generate personalized fitness plans.	High
4	Transactions processing	The system should accept user fitness details, submit a plan-generation request, receive the AI response, validate it, and present the generated plan to the user.	High
5	Reporting	The system should display workout plans, daily schedules, progress summaries, and basic fitness-plan history.	Medium
6	Business rules, etc.	Recommendations should consider fitness goal, fitness level, available time, preferences, and equipment; plans should include suitable exercises, rest, and progression guidance.	High
7	Compliance to laws or regulations	User data should be collected only as needed, protected from unauthorized access, and handled according to applicable privacy requirements and institutional policies.	Medium
8	Other	The system should allow users to provide feedback and use that feedback to	High

		support future plan adjustments.	
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Step 2: Non-Functional Requirements (NFR)

Non-functional requirements specify how well the FitBuddy system should operate.

S.No	NFR Category	Requirement Description / Target Metric or Acceptance Criteria
1	Performance & Speed	Normal pages should load within 2 seconds; a plan-generation request should provide a response within 10 seconds under normal network conditions.
2	Scalability	The design should support increasing users, stored plans, and progress records without major changes to the application architecture.
3	Security & Data Privacy	Use authenticated access, secure API credentials, input validation, and protected storage. Personal user data should not be exposed to unauthorized users.
4	Reliability & Availability	The application should handle temporary API/network failures gracefully and provide a clear retry or fallback message. Target availability: 99% during planned operating periods.
5	Usability & Accessibility	The interface should be simple, readable, mobile-friendly, and usable with clear labels, consistent navigation, and accessible controls.
6	Other - Maintainability & Portability	Use modular components and documented code so features can be updated easily; the application should be deployable on common web/server environments.

Requirement Summary

FitBuddy will provide an AI-assisted fitness planning service that accepts user requirements, uses Gemini models to generate personalized fitness plans, stores relevant profile and progress information, and presents clear workout guidance. The solution will prioritize personalization, security, usability, reliability, and maintainability.